Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

**MAT 2125, Winter 2023 – Midterm Exam** **Instructor: A. Tikuisis**

# Solutions

(Definitions and statements of theorems can be found in the course notes.)

Read the following instructions:

- The use of cellphones, electronic devices (including calculators), and course notes is strictly forbidden. All phones and electronic devices must be turned off and kept in your bags: do not leave them on you. If you are seen to have an electronic device on your person, we may ask you to leave the exam immediately, and fraud allegations could be made, which could lead to a mark of 0 (zero) on this midterm.
- The duration of this midterm is 75 minutes.
- This is a closed book midterm containing **5 questions** plus one bonus question.
- Do not detach the pages of this test.
- There is an additional blank page at the end of this exam that you may use as scrap paper. If you run out of space, you may use this page or the back of pages. Clearly indicate where to find your answer if it is not entirely contained in the space provided on the page.
- You must give clear and complete solutions, with calculations, explanations and justifications. Make sure that your answer is clearly indicated; you must convince me that you understand your solution in order to receive full marks.

**By signing below, you acknowledge that you are required to respect the above statements.**

*Signature*: _____

THIS SPACE IS RESERVED FOR THE MARKER:

| Question | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| Mark | | | | | | |
| Out of | 8 | 8 | 11 | 10 | 10 | 50 |

**1.** Label the following statements as true (T) or false (F). No justification is needed. (1 mark for each correct answer, 0 marks for an incorrect answer or no answer.) (8)

$\boxed{\text{F}}$ If $A \subseteq \mathbb{R}$ is a bounded subset, then $A$ contains a least upper bound.
*The set has a least upper bound, but it might not be contained in it; for example, consider* $A = (0, 1)$

$\boxed{\text{F}}$ If $(a_n)_{n=1}^{\infty}$ is a sequence that converges, then $(a_n)_{n=1}^{\infty}$ is monotone.
*Consider* $a_n := (-1)^n/n$ *for example.*

$\boxed{\text{T}}$ If $A \subseteq \mathbb{R}^d$ is an open set and $B \subseteq \mathbb{R}^d$ is a closed set, then $A \setminus B$ is an open set.
$\mathbb{R}^d \setminus B$ *is open, so* $A \setminus B = A \cap (\mathbb{R}^d \setminus B)$ *is also open.*

$\boxed{\text{T}}$ If $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}^d$ and we write $a_n = (a_n^{(1)}, \ldots, a_n^{(d)})$ for all $n$, then for $i = 1, \ldots, d$, the sequence $(a_n^{(i)})_{n=1}^{\infty}$ in $\mathbb{R}$ is Cauchy.
*This is true because* $|a_n^{(i)} - a_m^{(i)}| \leq \|a_n - a_m\|_2$.

$\boxed{\text{T}}$ There exists a set $A \subseteq \mathbb{R}$ such that $\overline{A} = \mathbb{R}$ and $\overline{(\mathbb{R} \setminus A)} = \mathbb{R}$.
*For example* $A = \mathbb{Q}$.

$\boxed{\text{F}}$ If $(a_n)_{n=1}^{\infty}$ is a sequence such that $a_n \geq 0$ for all $n$ and $a_n \to 0$ then by the Alternating Series Test, $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges.
*For example, if* $a_{2k} = 1/k$ *and* $a_m = 0$ *when* $m$ *is odd, then* $\sum_{n=1}^{\infty}(-1)^{n+1}a_n = \sum_{k=1}^{\infty} 1/k$*, which diverges.*

$\boxed{\text{T}}$ If $(a_n)_{n=1}^{\infty}$ has a subsequence that diverges then $(a_n)_{n=1}^{\infty}$ diverges.
*For example, if* $a_{2k} = 1/k$ *and* $a_m = 0$ *when* $m$ *is odd, then* $\sum_{n=1}^{\infty}(-1)^{n+1}a_n = \sum_{k=1}^{\infty} 1/k$*, which diverges.*

$\boxed{\text{T}}$ There exists a sequence $(a_n)_{n=1}^{\infty}$ such that $a_n \to 0$ and $\sum_{n=1}^{\infty} a_n$ diverges.
*For example, the Harmonic Series,* $\sum_{n=1}^{\infty} 1/n$*, diverges.*

**2.** Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.

(a) Define what it means for $(a_n)_{n=1}^{\infty}$ to **converge** to a number $L \in \mathbb{R}$. (3)

(b) Prove the **Squeeze Theorem**: if $(b_n)_{n=1}^{\infty}, (c_n)_{n=1}^{\infty}$ are other sequences, satisfying

$$a_n \leq b_n \leq c_n \quad \text{for all } n \in \mathbb{N}_{\geq 1},$$

and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$. (5)

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Solution: (a): See course notes.

(b): Let $\epsilon > 0$ be given. Since $a_n \to L$, there exists $n_{0,a}$, such that for all $n \geq n_{0,a}$,

$$L - \epsilon < a_n < L + \epsilon.$$

Since $c_n \to L$, there exists $n_{0,c}$, such that for all $n \geq n_{0,c}$,

$$L - \epsilon < c_n < L + \epsilon.$$

Set $n_0 := \max\{n_{0,a}, n_{0,c}\}$. Then for $n \geq n_0$,

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon,$$

as required.

**3.** (a) State the **Absolute Convergence Test** for a series, and state what it means for a series to converge **absolutely**. (3)

(b) Give an example of a series that converges, but that does not converge absolutely. Justify why your example has these properties. (4)

(c) Suppose that $(a_n)_{n=1}^{\infty}$ is a bounded sequence. Prove that the series

$$\sum_{n=1}^{\infty} \frac{na_n + 1}{n^3}$$

converges. (4)

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Solution: (a): See course notes.

(b): $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is an example. It converges by the Alternating Series Test. It does not converge absolutely, because when taking the series with the absolute values of the terms we get $\sum_{n=1}^{\infty} \frac{1}{n}$, which is the Harmonic Series, which does not converge.

(c): Let $M > 0$ be such that $|a_n| \leq M$ for all $n$. Then for $n \geq 1$ we have

$$\left| \frac{na_n + 1}{n^3} \right| \leq \frac{n|a_n| + n}{n^3} \leq \frac{M+1}{n^2}.$$

Since $\sum_{n=1}^{\infty} \frac{(M+1)}{n^2}$ converges, it follows by the Comparison Test that

$$\sum_{n=1}^{\infty} \left| \frac{na_n + 1}{n^3} \right|$$

converges. Therefore by the Absolute Convergence Test,

$$\sum_{n=1}^{\infty} \frac{na_n + 1}{n^3}$$

converges.

**4.** Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence of real numbers.

(a) Give a definition (or equivalent formulation) for

$$\limsup_{n\to\infty} a_n. \tag{3}$$

(b) Compute $\limsup_{n\to\infty} a_n$ where

$$a_n := \begin{cases} \frac{k+1}{k}, & \text{if } n = k! \text{ for some } k \in \mathbb{N}_{\geq 1}; \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

(c) Prove that if $(a_n)_{n=1}^{\infty}$ is a bounded sequence then there exists a subsequence $(a_{n_k})_{k=1}^{\infty}$ that converges to $\limsup_{n\to\infty} a_n$. (4)

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Solution: (a): Any of the following are acceptable:

$$\limsup_{n\to\infty} a_n = \inf\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \leq \beta \ \forall n \geq n_0\}.$$
$$\limsup_{n\to\infty} a_n = \inf_n \sup\{a_n, a_{n+1}, \dots\},$$
$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup\{a_n, a_{n+1}, \dots\}.$$

(b): The limsup is 1.

If $\beta > 1$ then there exists $k_0$ such that $\frac{k_0+1}{k_0} < \beta$. Consequently, if $n \geq k_0!$ then either $a_n = 0$ or $a_n = k!$ for some $k \geq k_0$. In the latter case, we get

$$a_n = \frac{k+1}{k} \leq \frac{k_0+1}{k_0} < \beta.$$

This shows that $\beta$ is an eventual upper bound. Hence, $\limsup a_n \leq 1$.

On the other hand, there are infinitely many $n$ for which $a_n \geq 1$ (namely, $n = k!$ for any $k \in \mathbb{N}_{\geq 1}$). Therefore, 1 is not an eventual upper bound, so $\limsup a_n \geq 1$.

(c): Set $L := \limsup a_n$. We will construct a subsequence $(a_{n_k})_{k=1}^{\infty}$ such that $|a_{n_k} - L| \leq \frac{1}{k}$ for all $k$. By the Squeeze Theorem, it will follow that $a_{n_k} \to L$.

For $k = 1$, we note that $L+1$ is an eventual upper bound so there is some $n_0$ such that $a_n \leq L+1$ for all $n \geq n_0$. We also note that $L-1$ is not an eventual

upper bound, so there exists $n_1 \geq n_0$ such that $a_{n_1} \geq L - 1$. Since $n_1 \geq n_0$, we also have $a_{n_1} \leq L + 1$, and thus $|a_{n_1} - L| \leq 1$.

Next for $k = 2$, $L + 1/2$ is an eventual upper bound, so there is some new $n_0$ such that $a_n \leq L + 1/2$ for all $n \geq n_0$. We also note that $L - 1/2$ is not an eventual upper bound, so there exists $n_2 \geq \max\{n_0, n_1\}$ such that $a_{n_2} \geq L - 1/2$.

Next for $k = 3$, $L + 1/3$ is an eventual upper bound, so there is some new $n_0$ such that $a_n \leq L + 1/3$ for all $n \geq n_0$. We also note that $L - 1/3$ is not an eventual upper bound, so there exists $n_3 \geq \max\{n_0, n_2\}$ such that $a_{n_3} \geq L - 1/3$.

Continuing in this way produces the desired subsequence.

**5.** (a) State the definition of what it means for a set $K \subseteq \mathbb{R}^d$ to be (sequentially) compact. (3)

(b) Prove that $K := \{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}$ is compact, from the definition. **Do not use the Heine–Borel theorem.** (5)

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Solution: (a): See course notes.

(b): Let $(a_n)_{n=1}^{\infty}$ be a sequence in $K$. Then $\|a_n\|_2 \leq 1$ for all $k$, so the sequence is bounded and thus by the Bolzano–Weierstrass Theorem, it has a subsequence $(a_{n_k})_{k=1}^{\infty}$ which converges to some $L \in \mathbb{R}^d$. It follows that $\|a_n\|_2 \to \|L\|_2$, and therefore $\|L\|_2 \leq 1$. Consequently, $L \in K$. This shows that $K$ is compact.

**6.** (Bonus) Let $(a_n)_{n=1}^{\infty}$ be a sequence that converges to $L$. For each $n \in \mathbb{N}_{\geq 1}$, define

$$b_n := \frac{a_1 + \cdots + a_n}{n}.$$

Prove that $(b_n)_{n=1}^{\infty}$ also converges to $L$. `(up to 5 bonus marks)`

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